CS5670: Computer Vision

Image transformations and image warping

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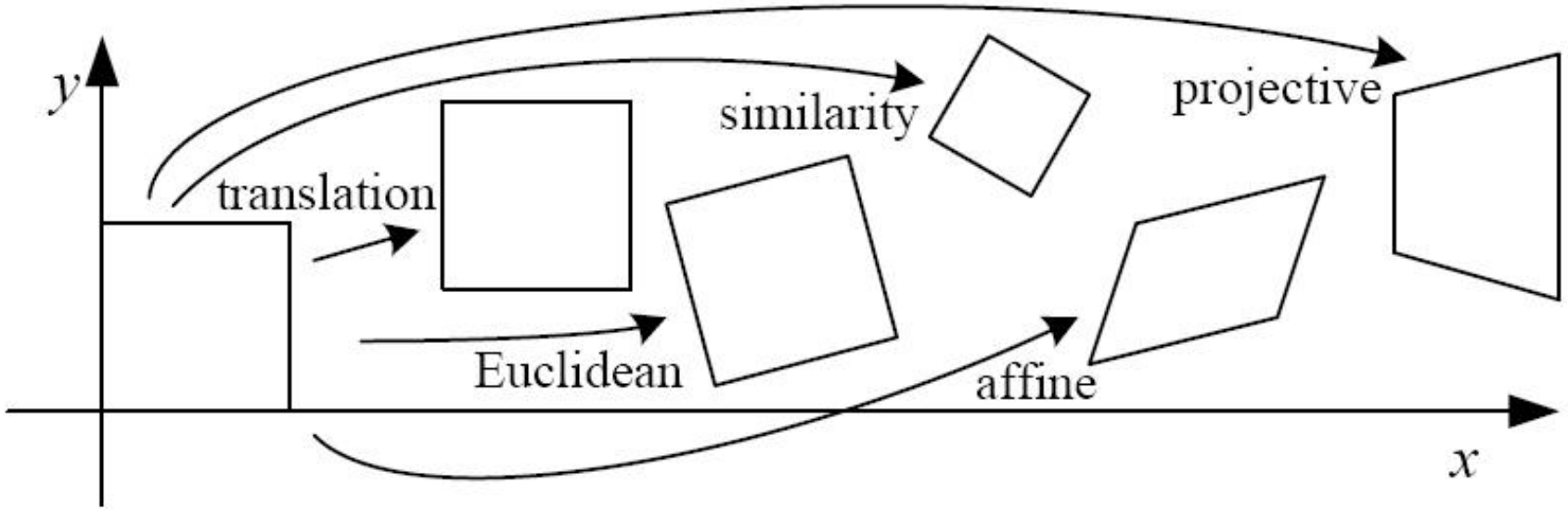


Image alignment



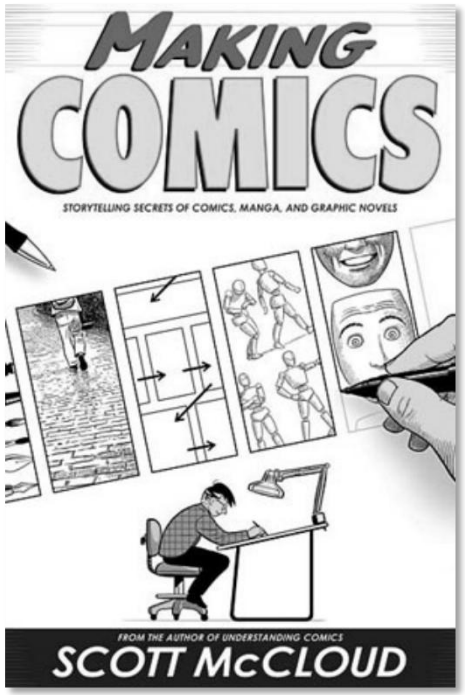


Image alignment



Why don’t these image line up exactly?

What is the geometric relationship between these two images?

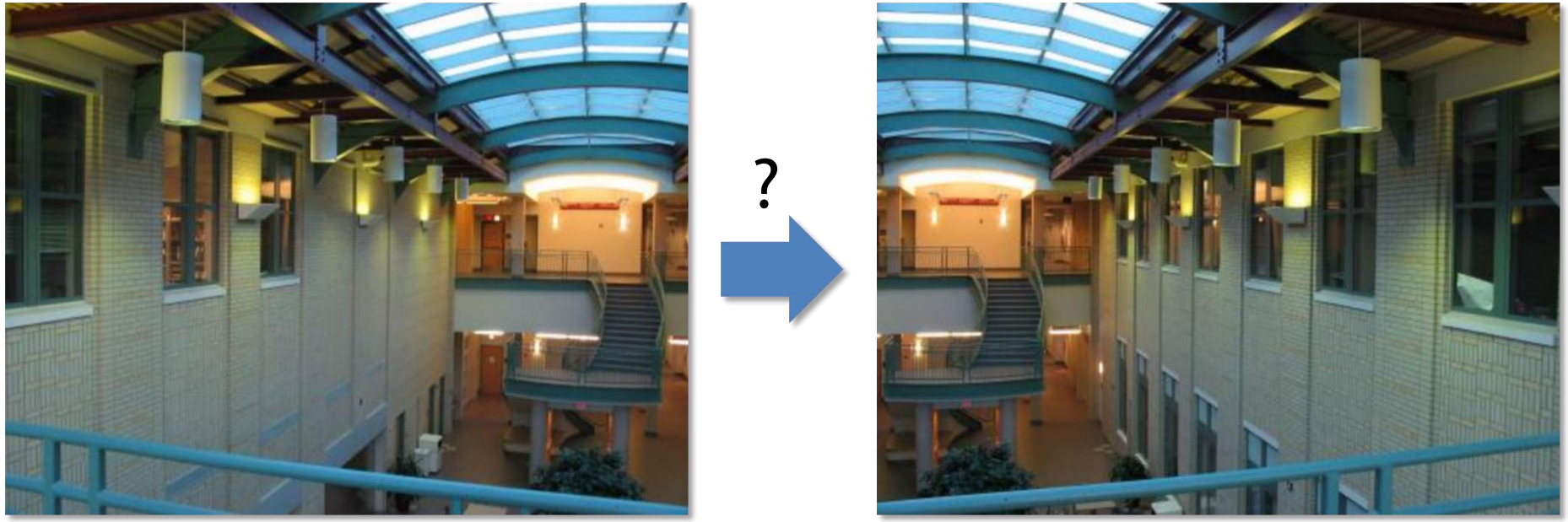


?



Answer: Similarity transformation (translation, rotation, uniform scale)

What is the geometric relationship between these two images?



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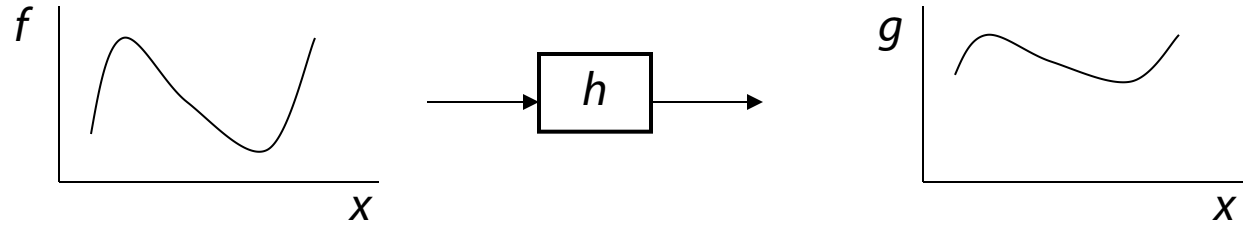
Very important for creating mosaics!

First, we need to know what this transformation is.

Second, we need to figure out how to compute it using feature matches.

Image Warping

- image filtering: change *range* of image
  - $g(x) = h(f(x))$



- image warping: change *domain* of image
  - $g(x) = f(h(x))$

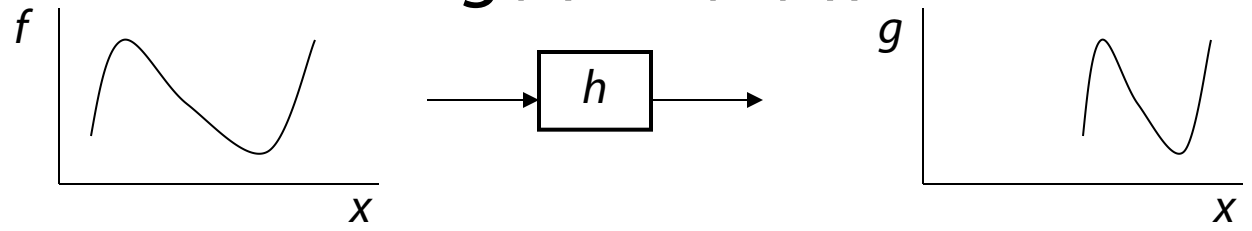
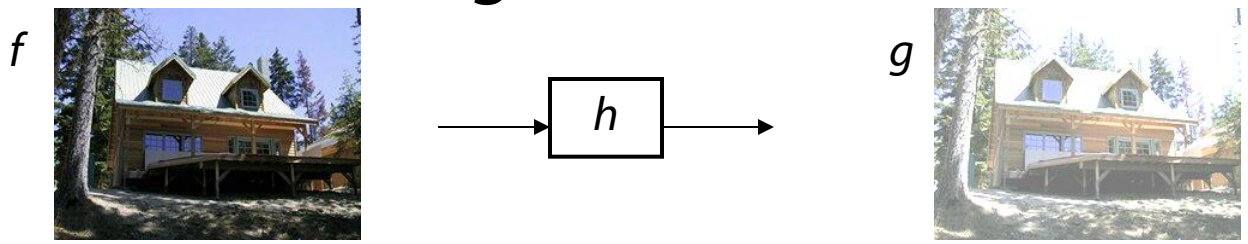


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